Execise 7

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Exercise 3.2

At the interface between the normal metal and the superconductor, Andreev reflection occur. The BdG equation can be written as:

$$\mathrm{i}\hbar\partial_t \begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} -\frac{\hbar^2}{2m}\nabla^2 - \mu(x) + V(x) & \Delta(x) \\ \Delta^* & \frac{\hbar^2}{2m}\nabla^2 + \mu(x) - V(x) \end{pmatrix} \begin{pmatrix} f \\ g \end{pmatrix}$$

where $\Delta(x) = \theta(x)\Delta$ and $V(x) = H\delta(x)$.

1. Prove $u_0^2 = 1 - v_0^2 = \frac{1}{2}(1 + \sqrt{E^2 - \Delta^2}/E)$ and $\hbar k^{\pm} = \sqrt{2m(\mu \pm \sqrt{E^2 - \Delta^2})}$.
2. Solve the scattering coefficient of this scattering problem.

Hint: use plane-wave ansatz:

$$\begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix} \mathrm{e}^{\mathrm{i}(kx - Et/\hbar)}$$

then solve the stationary eigenvalue equation and apply the boundary condition at $x = 0$.

Execise 3.3

Prove the current conservation of quasi-particles in Andreev reflection problem:

$$\partial_t \rho + \nabla \cdot \boldsymbol{J}_\rho = 0$$

where $\rho = |f|^2 + |g|^2$ and $\boldsymbol{J}_\rho = \frac{\hbar}{m}\left[\mathrm{Im}(f^*\nabla f) - \mathrm{Im}(g^*\nabla g)\right]$